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ORIGINAL ARTICLE

Active learning for Arabic sentiment analysis

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Abstract Sentiment analysis becomes an essential part of every social network, as it enables decision-makers to know more about users' opinions. Arabic sentiment analysis has a small number of datasets relative to English. This paper introduces active learning to the Arabic NLP community and demonstrates its power. Active learning is the process of selecting the data before labelling it to get the best benefits from the labelled data and improve performance. As it takes much time and effort to label large datasets, a better way is to label selected data and get higher performance. This paper applies active learning to the ArSarcasm-v2 dataset, which is labelled for sentiment, sarcasm, and dialect, by using the SAIDS "Sentiment Analysis Informed of Dialect and Sarcasm" model that predicts sarcasm and dialect and then uses them to predict the sentiment of the text. The paper runs multiple active learning experiments with different setups. The paper achieves state-of-the-art performance on ArSarcasm-v2 for sentiment analysis by achieving a 76.71 F1-score using 95% of the training dataset which means that adding some data points makes the performance worse. The paper also demonstrates that by using active learning, we achieve 97%, 99%, and 100% of the highest F1-score reached by all training data using only 10%, 20%, and 27% of the training data respectively. This means that using about one-quarter of the training data attains the same F1-score as using the full training data. As per our knowledge, this is the first paper to introduce active learning to the Arabic NLP community.

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1. Introduction

Active Learning is an approach to learning which introduces interactive data labelling. Instead of labelling all amounts of data and then training a model, the active learning process is to label a small amount of data then choose new data to be labelled and so on. Active learning focuses on choosing the best dataset that gives a desirable level of accuracy. In other words, active learning focuses on achieving a desirable level of accuracy using the smallest number of data points, thereby minimising the labelling cost [37].

It achieves high accuracy by selecting from a large pool of unlabelled data the most useful data points. Then gives these selected data points to the annotator to label them. By selecting which data points to be labelled, the annotation cost is reduced while the performance is on the same level [34].

On the other hand, the Arabic NLP community does not have large datasets. Due to the lack of Arabic annotated data, the average accuracy of Arabic NLP tasks is much lower than

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English tasks. As an example for the sentiment analysis task, which is one of the main tasks in NLP, there are a huge number of English datasets. Most of them have more than 100,000 data row but for the Arabic datasets, the average is 20,000 samples. Not only is the average lower, but also the number of available datasets is lower. In fact, the number of datasets available in English is hundreds of times more than the number of datasets available in Arabic [15]. Another example is sarcasm detection, there are less than ten Arabic datasets for sarcasm, on the other hand, there are hundreds of English datasets. The problem becomes more evident when different Arabic dialects are approached, there are far fewer datasets. Accordingly, active learning can be utilized to fill the gap between English and Arabic datasets available for the research community. Active learning is more attractive when the datasets are small, in this case, active learning secures a better way to utilize the data. In fact, the best practice is to use active learning while creating a new dataset. Actually, research papers use annotated datasets to demonstrate the efficiency of active learning. Datasets' creators should apply it while creating new datasets. The contributions offered by this work are:

- Introducing active learning to the Arabic NLP community by applying it to Arabic sentiment analysis.
- Experimenting with multiple parameters in active learning.
- Achieving state-of-the-art performance on ArSarcasm-v2 for sentiment analysis by attaining 76.71 F1-score using 95% of the training dataset.
- Achieving 97%, 99%, and 100% of the highest F1-score reached by training on the whole dataset using only 10%, 20%, and 27% of the dataset respectively.

This paper is organized as follows, Section 2 shows the related work on active learning and sentiment analysis. Section 3 describes the dataset used in this work and shows some data statistics. Section 4 elaborates on the active learning main approach, used query strategies, and used models. Section 5 shows the experiments' results and finally, Section 6 concludes the work.

2. Related work

This paper works on Arabic sentiment analysis tasks using active learning for selecting the data for training. In this section, the existing methods for sentiment analysis and active learning are discussed.

2.1. Active learning

Active learning is a machine learning approach that involves the model actively selecting informative examples from the unlabeled data for labeling to become training data. This method is particularly relevant in deep learning and machine learning due to the high cost and time associated with labeling large datasets [34]. It started even before deep learning [27], the idea itself was formulated at that time. Using active learning with deep learning started at the beginning of using deep learning [37,41].

Active learning differs from passive learning, which follows the conventional approach of building a large dataset and then training a model to fit it. In active learning, the dataset is not constructed all at once; instead, the data selection for labeling is performed iteratively. This selection process, known as "Query Data", involves using a query strategy to choose the most informative examples for labeling. Consequently, active learning methods showcase the capability to improve model performance, minimize annotation expenses, and effectively handle complex data distributions.

One popular active learning method is "Uncertainty Sampling" [32], which selects samples based on the model's prediction uncertainty. Although it is simple to implement, its performance may suffer when uncertainty estimates are unreliable or when multiple classes have similar confidence levels. Another method is "Query by Committee" [38], which utilizes a committee of models to identify uncertain samples, enhancing sample diversity. However, forming and maintaining the committee can be computationally expensive. "Density-Weighted Active Learning" [26] samples data based on their density in the feature space, improving performance on sparse or imbalanced data. However, it requires a density estimator, which can be computationally expensive. "Expected Model Change" [31] selects samples with the potential to change the model's predictions, refining its confidence. It involves a computationally expensive optimization process. Lastly, "Active Learning with Deep Generative Models" [40] employs deep generative models to select samples, capturing complex data distributions. However, training on large datasets may be challenging due to resource requirements.

This paper presents a novel application of active learning in Arabic Natural Language Processing (NLP), which is the first known instance of active learning being utilized in this domain. While active learning has been previously applied in various languages such as English, French, Spanish, German, Dutch, Japanese [29], Chinese [33], Hindi, Indonesian [12], Danish, Greek, Italian, Portuguese, and Swedish [18], its application in Arabic NLP has not been explored until now.

2.2. Sentiment analysis

Sentiment analysis, also known as opinion analysis or opinion mining, has gained widespread acceptance in various fields [39]. It plays a crucial role in automatically analyzing user-generated data and monitoring public opinion, particularly with the rise of the Internet. As a subfield of natural language processing (NLP), sentiment analysis focuses on extracting sentiments from the text. The field has witnessed significant growth, with researchers presenting frameworks [11] and methodologies for processing unstructured data and identifying sentiments. Machine learning techniques, such as detecting opinion spam and fraudulent reviews [13], have been explored, along with the analysis of internet reviews and the application of machine learning algorithms to social media analysis [24].

Sentiment analysis has been studied across multiple levels, including Document Level, Sentence Level, Phrase Level, and Aspect Level. Document-level sentiment analysis assigns a single polarity to an entire document, categorizing it as positive, negative, or neutral. It is less commonly used but can be applied to classify chapters or pages of a book. Both supervised and unsupervised learning methods are utilized, with domain-specific and cross-language sentiment analysis posing significant challenges [10].

The sentence-level sentiment analysis focuses on analyzing each sentence individually, determining its polarity. This approach is useful when a document contains diverse sentiments. Each sentence's polarity can be determined independently, similar to document-level analysis but with more data and computational resources [43]. Aggregating sentence polarities can provide an overall sentiment for the document. The sentence-level analysis becomes crucial for challenging tasks such as conditional sentences or ambiguous statements [19].

Phrase-level sentiment analysis involves classifying opinion words at the phrase level. It is commonly used in product reviews, where a single aspect is expressed in a phrase. This approach has gained attention from researchers. The sentence-level analysis is more beneficial as it captures both positive and negative statements. Words, particularly adjectives, play a fundamental role in determining subjectivity. The choice of words reflects demographic characteristics, desires, social status, and other psychological and social attributes [20].

2.3. Arabic sentiment analysis

In contrast to English sentiment analysis, which benefits from a wide range of over ten thousand related datasets [28], Arabic sentiment analysis lags behind with only hundreds of available datasets. Early work on Arabic sentiment analysis began with the labeling of reviews as positive or negative by [1]. Since then, researchers have developed more sophisticated methods for Arabic sentiment analysis, initially focusing on Modern Standard Arabic (MSA) and later expanding their scope to include Arabic dialects [30,24].

Despite the limited dataset availability, there are over fifty datasets dedicated to Arabic sentiment analysis, including datasets such as [17,23,25,35,16,22]. Researchers have utilized these datasets to address sentiment analysis using various system approaches [4,8,14]. A comparative study on Arabic sentiment analysis [6] categorizes the work into three clusters: classical machine learning models, deep learning models with word embeddings, and transformer-based models. Overall, the transformer-based models, including AraBERT [9], ARBERT and MARBERT [3], demonstrate higher accuracy compared to the other two clusters.

ARBERT and MARBERT are state-of-the-art Arabic language models based on the BERT architecture. ARBERT is pre-trained on Modern Standard Arabic text data, while MARBERT is pre-trained on both Modern Standard Arabic and Arabic dialects. The training data for both models is collected from diverse sources, including news articles, Wikipedia, and scientific papers. These large models, with ARBERT having 125 million parameters and MARBERT having 340 million parameters, excel in various natural language processing tasks such as text classification and named entity recognition. However, their size makes them computationally demanding, requiring powerful hardware for efficient training and inference. The MARBERT model specifically focuses on understanding the Arabic social media language, while ARBERT is trained on Modern Standard Arabic. Both models can be fine-tuned for specific tasks with limited additional data, showcasing their adaptability and performance across multiple NLP applications.

3. Dataset

The dataset used in this work is ArSarcasm-v2 [7] which is an Arabic dataset that was released at the WANLP conference in 2021. The dataset consists of 15 thousand tweets including 3 thousand tweets for testing. The same test set was used as released on the original dataset. The dataset is labelled for sentiment, sarcasm, and dialect. The sentiment classes are negative, neutral, and positive, the sarcasm classes are sarcastic and non-sarcastic, and the dialects are MSA, Egypt, Levantine, Maghreb, and Gulf. Crowd-sourcing platforms were used to annotate the dataset. The tweets are the combination of the ArSarcasm dataset [5], DAICT dataset [2], and random tweets collected by the authors of the dataset [7]. The ArSarcasm-v2 dataset has 12,548 tweets for training and 3000 tweets for testing. Table 1 shows the distribution of tweets between the different class labels in the training data. Most of the data is labelled as neutral, non-sarcastic, and MSA in sentiment, sarcasm, and dialect respectively.

The cross-tabulation between sentiment analysis labels and both sarcasm detection and dialect identification labels can be shown in Table 2. The relation between sentiment and sarcasm shows that more than 90 percent of the tweets labelled sarcastic are labelled negative in sentiment analysis. Also, the sentiment-dialect relation shows that most of the MSA tweets are labelled neutral, and more than 50 percent of the Egyptian tweets are labelled negative. As the number of North African tweets is only 43 tweets, it was removed from the table.

Table 3 demonstrates the percentage of sarcastic tweets over the total number of tweets for each dialect. As shown in the table, Egyptian tweets in this dataset are the most sarcastic. The NOR tweets were also removed from this table as they are 43 tweets only.

Although the ArSarcasm-v2 dataset is labeled for sentiment, sarcasm, and dialect, our main focus was on sentiment analysis. Therefore, no special preprocessing or modifications were made to the dataset in terms of handling data balancing. The dataset is already well-balanced, with a sufficient number of data points in each sentiment class. One advantage of working with the ArSarcasm-v2 dataset is its balanced distribution of sentiment labels. Table 1 provides an overview of the distribution of tweets among the different sentiment classes in the training data. It can be observed that the dataset includes a reasonable number of positive, neutral, and negative sentiment

Table 1 Distribution of the different class labels for sentiment, sarcasm, and dialect for the ArSarcasm-v2 dataset.

Task	Label	Count
Sentiment	Positive	2,180
	Neutral	5,747
	Negative	4,621
Sarcasm	Sarcastic	2,168
	Non-sarcastic	10,380
Dialect	MSA	8,562
	EGY	2,675
	Gulf	644
	LEV	624
	NOR	43
Total		12,548

Table 2 Cross-tabulation between sentiment and both sarcasm and dialect labels for the ArSarcasm-v2 dataset.

Task	Label	Sentiment analysis		
		POS	NEU	NEG
Sarcasm	Sarcastic	58	171	1,939
	Non-sarcastic	2,122	5,576	2,682
Dialect	MSA	1,405	4,486	2,671
	EGY	506	793	1,376
	Gulf	121	259	264
	LEV	142	197	285

Table 3 Percentage of sarcastic tweets for each dialect for the ArSarcasm-v2 dataset.

Dialect	Sarcasm percentage
MSA	10.83 %
EGY	34.77 %
Gulf	24.38 %
LEV	22.12 %

tweets, allowing for comprehensive sentiment analysis. This balanced nature of the dataset minimizes potential biases and enables reliable modeling of sentiment.

Additionally, the ArSarcasm-v2 dataset was constructed by combining data from various sources, including the DAICT dataset and additional random tweets. This utilization of multiple sources ensures the diversity and richness of the dataset, capturing various linguistic styles and expressions prevalent in Arabic social media. Overall, the nature of the ArSarcasm-v2 dataset provides a solid foundation for our work on the sentiment analysis. While our primary focus was sentiment, the dataset remains a valuable resource for future investigations into sarcasm detection, dialect identification, and other related tasks in Arabic natural language processing.

4. Methodology

This section presents a detailed description of the active learning methodology with the ArSarcasm-v2 dataset.

4.1. Active learning approach

The main active learning approach is presented in Fig. 1, which shows the general cycle of active learning. It starts with an unlabelled data pool that has a large amount of data. Collecting massive unlabelled data now is not an issue at all. Then after querying some unlabelled data, an annotator takes the data, labels it, and adds it to the training set. Then train a new model using the labelled dataset. This is the main cycle that is described for active learning [37].

This approach can be outlined by the pseudo-code in Algorithm 1. In the proposed work, the same main approach was used with slight changes to match the case. The input dataset was the ArSarcasm-v2. While running, some labels are made available to the model and the others are unavailable and seen as unlabelled. The "initial data" percentage was 1 percent in some experiments and 5 in others. Steps 2 and 7 are performed by moving the selected data from the unlabelled partition to the labelled one and using the label which is originally obtained from the ArSarcasm-v2 dataset. In step 4, the base model which is described in Section 4.3 is trained from scratch using the labelled data only. In step 5, a normal inference is done using the trained model to all the labelled and unlabelled data. Step 6 selects from the unlabelled data some data that will benefit the model if it was labelled. This is done by some query strategies that are discussed in Section 4.2. In step 9, "satisfaction" is achieved in this work when the full ArSarcasm-v2 dataset is labelled or when the accuracy of the model is as expected.

Algorithm 1. Active Learning Algorithm

Input: Large unlabelled dataset
1: Choose randomly "initial data" from unlabelled data
2: Label the "initial data"
3: **repeat**
4: Train a new model using the training set
5: Using the trained model, evaluate all data
6: Using query strategy, select some data to annotate
7: Annotate the queried data
8: Add annotated data to the training set
9: **until** satisfaction

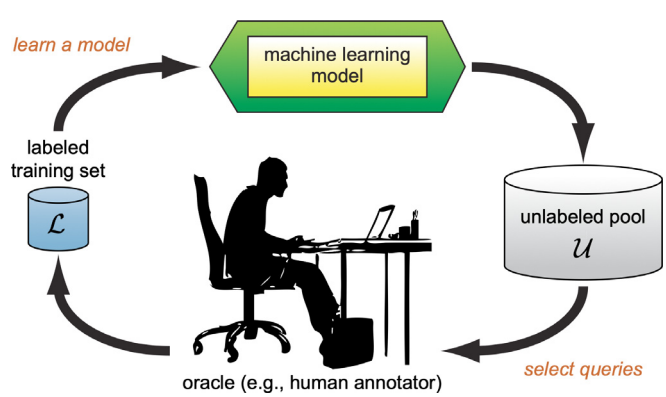


Fig. 1 Active Learning Cycle [37].

4.2. Query strategies

In this paper, multiple active learning query strategies were used, and this subsection describes them. Five query strategies were used. Three out of a total of five query strategies used in this research are Least Confidence sampling, Margin sampling, and Entropy sampling which were introduced in the first paper that applied active learning with deep learning [41]. The other two used strategies are Core-set sampling and K-means sampling [36]. Each query strategy is given the whole data, the model, and the query size (N) at each step in the loop and should output specific data points to be labelled.

Least Confidence Sampling Strategy This strategy uses the model to infer the unlabelled data and gets the prediction probability for each data point. It selects the N data points that have the lowest prediction probability to be labelled. This involves the probability of the highest class only in the selection process. Eq. 1 is for selecting one data point x by taking the lowest probability after taking the maximum class probability for each data point.

$$x = argmin_x \; max_c \; P(C_i|x_j) \tag{1}$$

Margin Sampling Strategy This strategy uses the model to infer the unlabelled data and gets the prediction probability for each data point also. It selects the N data points that have the lowest margin of probability between the top predicted class and the second predicted class. This involves the probability of the highest two classes in the selection process. Eq. 2 is for selecting one data point x by taking the data point with the lowest difference between the two highest probability classes.

$$x = argmin_x(max_c \; P(C_i|x_j) - 2^{ed} max_c \; P(C_i|x_j)) \tag{2}$$

Entropy Sampling Strategy This strategy uses the model to infer unlabelled data and gets the prediction probability for each data point also. It selects the N data points that have the highest class prediction information entropy. This involves the probability of all classes in the selection process. Eq. 3 is for selecting one data point x by calculating the entropy from the class prediction and taking the maximum data point.

$$x = argmax_x \; -\sum_c P(C_i|x_j) * \log P(C_i|x_j) \tag{3}$$

K-means Sampling Strategy This strategy uses the model to infer unlabelled data and gets the embedding of the input. This strategy can be outlined by the pseudo-code in Algorithm 2. The embedding of the input is the output of the last layer of the transformer-based model. In the normal transformer-based model, the embedding size is equal to 768, but due to computing limitations, PCA was applied to project it to a space of size 128. Then it makes a K-means cluster with k equal to N. Then it selects the centroids of each cluster to be the queried data points. This strategy does not involve the probabilities but it involves the embedding of the input.

Algorithm 2. K-means Sampling Strategy Algorithm

Input: Trained Model, Unlabelled Data, N
Output: N queried row
Use the trained model to do inference to all unlabelled data
Extract sentence embedding for each row
Project embeddings to space of size 128 using PCA
Cluster the embeddings with K-means and k = N
Initialize queried rows
for all clusters **do**
    Select the nearest row to the cluster center
    Add it to query rows
end for
return Queried rows

Core-set Sampling Strategy This strategy uses the model to infer all labelled and unlabelled data and get the embedding of the inputs. Algorithm 3 outlines the strategy. The embedding is exactly the same as the K-means strategy. Then it takes the farthest unlabelled point from the labelled ones. Then it iterates until it selects N points. This strategy does not involve the probabilities but it involves the embedding of the labelled and unlabelled data. Also, it is the only strategy that inferences the labelled dataset in each step.

Algorithm 3. Core-set Sampling Strategy Algorithm

Input: Model, labelled Data D_l, Unlabelled Data D_u, N
Output: N queried row
for D in $[D_l, D_u]$ **do**
    Use the trained model to do inference D
    Extract sentence embedding for each row
    Project embeddings to space of size 128 using PCA
    D = projected embeddings
end for
Initialize S the selected rows
for i = 0 To N **do**
    $u = argmax_{i \subset D_u}(min_{j \subset D_l}(\|x_i - x_j\|))$
    $S = S \cup \{u\}$
    $D_l = D_l \cup \{u\}$
    $D_u = D_u - \{u\}$
end for
return S

4.3. Model architecture

This subsection describes the used model architecture in the active learning loop. The model architecture was named SAIDS [21]. Fig. 2 shows the SAIDS model architecture. It passes the input text to the MARBERTv2 pretrained model to generate sentence embedding. Then, it passes the sentence embedding to the "Sarcasm Model" and "Dialect Model" to predict the sarcasm and dialect of the text. Then it passes the sentence embedding and the outputs of the "Sarcasm Model"

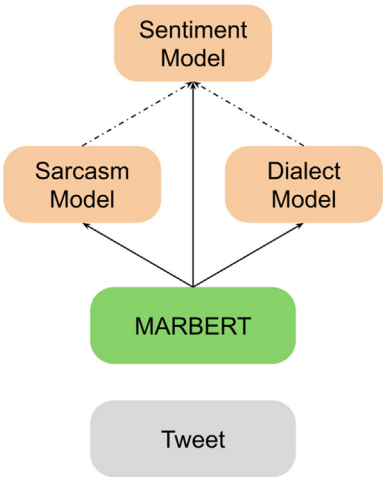


Fig. 2 SAIDS Model Architecture.

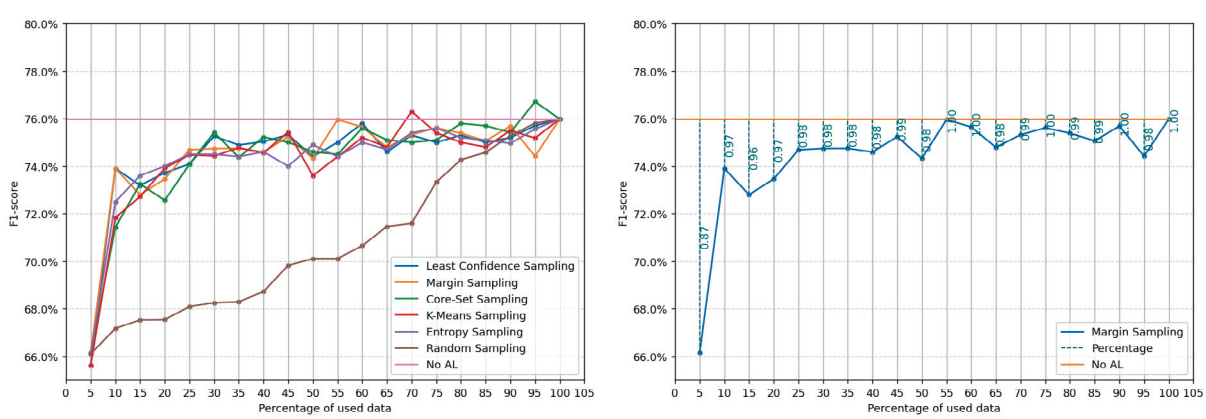
and "Dialect Model" to the "Sentiment Model" that predicts the sentiment of the text. This means that SAIDS uses the prediction of sarcasm to support the prediction of sentiment.

There were multiple setups offered with SAIDS [21], for module architecture, a neural network with one classification layer and without hidden layers was used. For the backpropagation path, there was the option of making the loss of sentiment analysis model pass through the "Sarcasm Model" and "Dialect Model" or not. It was proved experimentally that it is better to make the sentiment loss not propagate through the "Sarcasm Model" and the "Dialect Model" so this setup was used. Softmax was used as an activation function to the output of the "Sarcasm Model" and "Dialect Model" before passing them to the "Sentiment Model". For task-by-task training, although the original paper recommends not to train all tasks together, the same accuracy was achieved by training the full system together. The MARBERTv2 model provided by HuggingFace's transformers library [42] was used. For each loop, the models were trained from scratch and trained for five epochs with a learning rate $1e^{-5}$using the Adam optimizer. Tweets that have sequence lengths greater than 128 tokens were truncated.

5. Experiments and results

In this section, the different experiments are illustrated and their results are discussed. The parameters that were changed over the experiments are the maximum training data size, initial labelled data size, and query size. All five query strategies were used in all the experiments, in addition to another strategy which is the random query strategy. The random query strategy is the strategy of adding N data points randomly from the unlabelled data points, it is used for the baseline. Another baseline is the "No Active Learning" ("No AL") baseline which uses the full data to train the model. The training was done using the SAIDS model, which means that the model learns sentiment, sarcasm, and dialect but the main concern here is sentiment analysis so the query strategy works on it. The reported results are all for sentiment analysis using the average F1-score for positive and negative classes only. **A. Experiment One** The parameters used in this experiment are: maximum training data size equals 12,548 (100%) tweets, initial labelled data size equals 627 (5%) tweets, and query size equals 627 (5%) tweets. Fig. 3 is divided into two figures, Fig. 3a shows the F1-score for the sentiment analysis for each data percentage (e.g. initial data is 5% the accuracy is about 66% then after one query the labelled data is 10% and the accuracies are around 74%). It demonstrates that using a known query strategy is much better than using random sampling, while using 30% of the data, query strategies get 99% of the "No AL" F1-score, the random sampling gets only 79% of the "No AL" F1-score. It also shows that using Least Confidence sampling achieves the same F1-score as "No AL" while using only 55% of the data. Moreover, the K-means sampling outperforms the "No AL" while using 70% of the data. Regarding the Core-set sampling, it outperforms the "No AL" while using 95% of the data by achieving a 76.71 F1-score. Accordingly, it is concluded that some data points can make performance degradation for the model instead of improving it, so active learning is necessary even if you are not the creator of the dataset to remove bad data points.

And Fig. 3b shows the F1-score for sentiment analysis using the Margin query strategy only, it also shows the performance gap between "No AL" which is trained using the full



(a) F1-score for each data percentage with five query strategies and two baselines.

(b) F1-score for each data percentage with the Margin query strategy and "No AL" baseline.

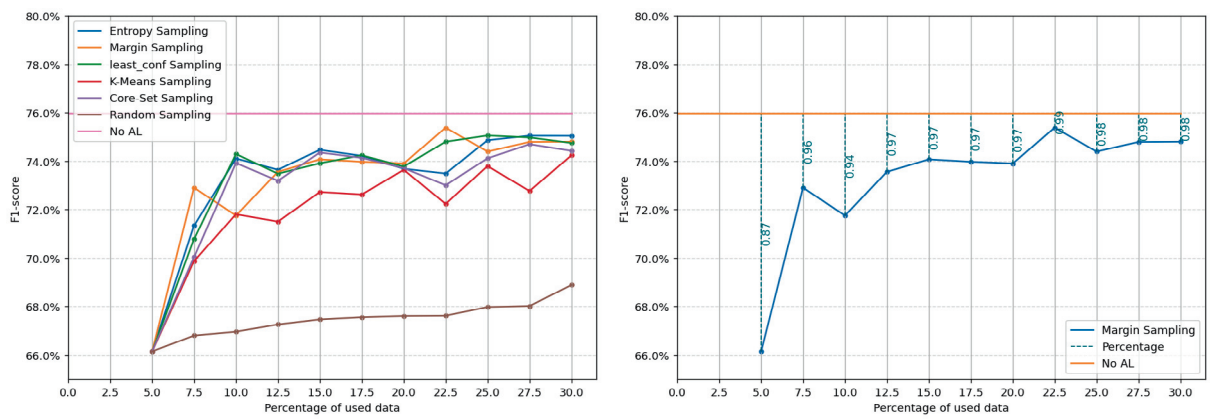
Fig. 3 Sentiment analysis performance for the first experiment.

data and the Margin query strategy which selects a percentage of the data to train on. It focuses on the Margin strategy curve as this is the best strategy concerning the earliest high score. Fig. 3b shows that the model can reach 97% of the full accuracy using only 10% of the dataset. And it reaches 99% of "No AL" using only 30% of the data. Also, by using 60% of the data it reaches the same F1-score as "No AL". This supports the main idea that a dataset creator should use active learning while creating it to enable researchers to get the highest accuracy.

B. Experiment Two The parameters used in this experiment are: maximum training data size equals 3,765 (30%) tweets, initial labelled data size equals 627 (5%) tweets, and query size equals 314 (2.5%) tweets. Fig. 4 is divided into two figures, Fig. 4a shows the F1-score for the sentiment analysis for each data percentage. It also demonstrates that using a known query strategy is much better than using random sampling. Moreover, it shows that using most of the query strategies gets similar F1-scores even if they do not query the same data points, so dataset creators can use the most simple active learning strategy and they would still get better results.

And Fig. 4b shows F1-score for sentiment analysis using the Margin query strategy only, it also shows the performance gap between "No AL" which is trained using the full data and the Margin query strategy which selects a percentage of the data to train on. It focuses on the Margin strategy curve as this is the best strategy concerning the earliest high score. It is clear that the model can reach 96% of the full accuracy using only 7.5% of the dataset. And it reaches 99% of "No AL" using only 22.5% of the data which is less than 30%, the percentage needed for the first experiment. The second experiment demonstrates that using a smaller query size achieves a higher F1-score with the same data size. In other words, smaller data can be used to reach 99% of the F1-score by using a smaller query size for selecting this data.

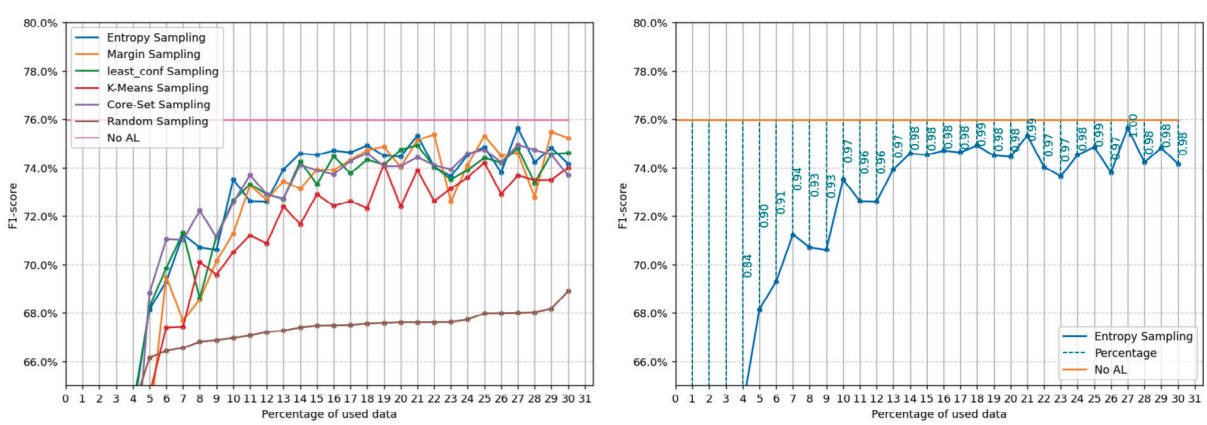
C. Experiment Three The parameters used in this experiment are: maximum training data size equals 3,765 (30%) tweets, initial labelled data size equals 125 (1%) tweets, and query size equals 125 (1%) tweets. Fig. 5 is divided into two figures, Fig. 5a shows the F1-score for the sentiment analysis for each data percentage. It also demonstrates the same regarding random sampling. The experiment started using



(a) F1-score for each data percentage with five query strategies and two baselines.

(b) F1-score for each data percentage with the Margin query strategy and "No AL" baseline.

Fig. 4 Sentiment analysis performance for the second experiment.



(a) F1-score for each data percentage with five query strategies and two baselines.

(b) F1-score for each data percentage with the Entropy query strategy and "No AL" baseline.

Fig. 5 Sentiment analysis performance for the third experiment.

1% of the data so the starting F1-score was too low, which is why it was not shown in the figure. It also shows that using most of the query strategies gets similar F1-scores even if they do not query the same data points, but the Entropy sampling performs better at the first queries.

And Fig. 5b shows F1-score for sentiment analysis using the Entropy query strategy only, it also shows the performance gap between "No AL" which is trained using the full data and the Entropy query strategy that selects a percentage of the data to train on. It focuses on the Entropy strategy curve which achieves the highest F1-score at most of the points. It is clear that the model can reach 90% of the full accuracy using only 5% of the dataset. And it reaches 99% of "No AL" using only 21% of the data and reaches the same "No AL" F1-score using 27% of the data only. It is incredible to reach the same F1-score using only a quarter of the data. The third experiment also shows that a smaller query size is better in terms of accuracy.

Query Size Comparison Each experiment used a different query size while running. Fig. 6 shows the curves of the average F1-score over query strategies in each experiment. It demonstrates that using a smaller query size selects better data points that achieve a higher F1-score. It can be shown that a query size equal to 1% is better than the others on almost all training data sizes, also a query size equal to 2.5% is better than 5%. On the other hand, reducing the query size increases the computational cost, as the model should be trained from scratch after each query. Also, the huge reduction in the query size will not affect the model training, for example, if the query size is one data point at a time, the model will not be affected.

Cost Analysis In this paper, the trade-off between computational costs (model training) and labeling (annotation) costs is considered, highlighting the importance of comparing both aspects. It should be noted that the dataset used was already annotated but we can make an estimate of the annotation cost, if the data was annotated using the "labelbox" website, the annotation cost for the full dataset of 12,000 sentences for one task would be approximately 83 USD. Since our dataset includes multiple labels (sentiment, sarcasm, and dialect), the annotation cost amounts to around 250 USD as of today. It's worth mentioning that the annotation cost is proportional to the data size. On the other hand, for the computational costs, we used Google Colab to train our models. Training the full dataset takes approximately 16 min using a T4 GPU, which costs around 0.068 USD for one complete training session. Similarly, the computational cost will be proportional to the training data size.

In the passive learning method, which has been commonly used in Arabic NLP, the cost would be the sum of the annotation cost for the full dataset and training the model once, resulting in a total cost of 250.068 USD. Conversely, the active learning method provides the creators with the flexibility to choose the desired accuracy level. If the creators are satisfied with a 73.70 F1-score, which is 97% of the maximum F1-score achieved with the full dataset in experiment one, this accuracy is achieved by only 10% of the full data. The cost then is 10% of the cost of the full dataset $250 * 0.10 = 25$, and the cost of training the model twice, first using 5% of the data and then using 10% of the data $(0.068 * 0.05) + (0.068 * 0.10) = 0.01$. Consequently, the total cost would be 25.01 USD. In the same experiment, experiment one, where an accuracy of 99% was achieved using 30% of the data, the cost of annotating 30% of the data would be 75 USD, and the cost of training the model six times would be 0.2 USD. Therefore, the total cost would be 75.2 USD. This indicates that the main cost is attributed to annotation, and reducing the number of annotated data points would effectively reduce the overall cost. Thus, using active learning to

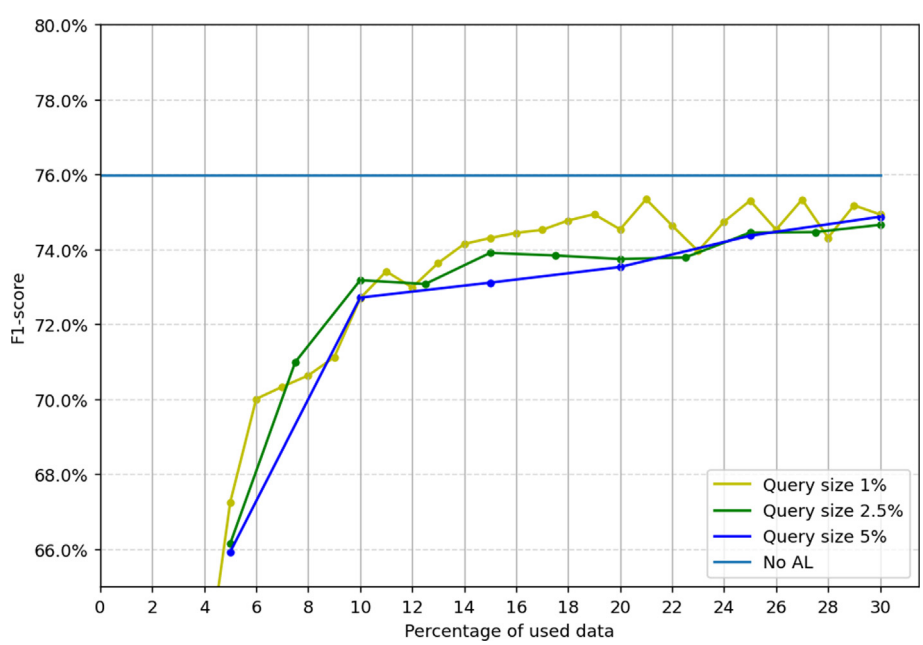


Fig. 6 F1-score for sentiment analysis for each data percentage for different query sizes and "No AL" baseline.

minimize costs is no longer an option but a necessity for every company or researcher.

6. Conclusion

Sentiment analysis systems are very important as they are used extensively in decision-making, however, Arabic sentiment analysis has a relatively small number of datasets and a small number of rows in each dataset. This work proves the effectiveness of active learning while creating a dataset and training new models. Active learning is the process of selecting the data that should be labelled from a massive number of unlabelled data points. Selecting the data comes with different query strategies. In this work, active learning was used while training on the ArSarcasm-v2 dataset which is labelled for sentiment, sarcasm, and dialect. The trained model architecture follows SAIDS architecture. SAIDS model predicts sentiment, sarcasm, and dialect and uses its prediction of sarcasm and dialect to enhance its prediction of sentiment. The state-of-the-art performance was achieved on the ArSarcasm-v2 dataset for sentiment analysis by attaining a 76.71 F1-score using 95% of the training dataset. Three active learning experiments were run, each one has a different query size, runs with five query strategies and the random baseline which represents passive learning. Any query strategy was proved much better than the random baseline by a large margin which supports the necessity of using active learning while creating datasets. From the three experiments, the model achieves 97%, 99%, and 100% of the highest F1-score reached by all training data using only 10%, 20%, and 27% of the training data respectively. Also, the model achieves 101% of the highest F1-score reached by all training data using 70% of the training data which means that by using selected 70% of the training data the model achieves a higher F1-score than training by the whole data. It also means that adding some data points to the dataset may even degrade the performance. To the best of our knowledge, this work is the first to introduce active learning to the Arabic NLP community despite its successful utilization in various widely spoken languages such as English, German, and Spanish. By utilizing active learning, better results can be achieved using significantly less training data, which aligns perfectly with the needs of the Arabic NLP community. Given the existing shortage of data for several tasks in Arabic NLP, the adoption of active learning holds great potential for overcoming this challenge and improving the performance of sentiment analysis systems.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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